```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
```

```
def predict(X, weights, lambdaa = 1):
  net = np.dot(X, weights)
  return (1 / (1 + np.exp(-lambdaa * net)))

def calculate_dw(X, y, y_pred):
  return ((y_pred - y) * (y_pred) * (1 - y_pred) * X)

def calculate_error(y, y_pred):
  return np.square(y_pred - y)
```

```
def train_vanilla(X, y, weights, epochs = 500, lr = 0.001, gamma = 0.9):
    upd = 0
    for epoch in range(epochs):
        dw = 0
        error = 0

        for (Xi, yi) in zip(X, y):
            y_pred = predict(Xi, weights)
            dw += calculate_dw(Xi, yi, y_pred)
            error += calculate_error(yi, y_pred)

        upd = (lr * dw) + (gamma * upd)
        weights -= upd
        error /= (2 * len(X))

        if((epoch + 1) % 50 == 0):
            print(f"Weights after {epoch + 1} epochs: {weights} \n")
            print(f"Error after {epoch + 1} epochs: {error} \n")
```

```
def train_stochastic(X, y, weights, epochs = 500, lr = 0.001, gamma = 0.9):
    upd = 0

    for epoch in range(epochs):
        dw = 0
        error = 0

        for (Xi, yi) in zip(X, y):
            y_pred = predict(Xi, weights)
            dw += calculate_dw(Xi, yi, y_pred)
            error += calculate_error(yi, y_pred)
            upd = (lr * dw) + (gamma * upd)
            weights -= upd

        error /= (2 * len(X))

        if((epoch + 1) % 50 == 0):
            print(f"Weights after {epoch + 1} epochs: {weights} \n")
            print(f"Error after {epoch + 1} epochs: {error} \n")
```

```
def train_minibatch(X, y, weights, epochs = 500, lr = 0.001, gamma = 0.9, bs = 32):
    upd = 0

    for epoch in range(epochs):
        dw = 0
        error = 0
        i = 0

        for (Xi, yi) in zip(X, y):
            y_pred = predict(Xi, weights)
            dw += calculate_dw(Xi, yi, y_pred)
            i += 1
            error += calculate_error(yi, y_pred)

            if i % bs == 0 or i == len(X):
                upd = (lr * dw) + (gamma * upd)
                weights -= upd
                dw = 0

        error /= (2 * len(X))

        if((epoch + 1) % 100 == 1):
            print(f"Weights after {epoch + 1} epochs: {weights} \n")
            print(f"Error after {epoch + 1} epochs: {error} \n")
```

```
df = pd.read_csv('bank_note.csv')
df.insert(4, 'x0', 1)
```

```
X = df[['variance', 'skewness', 'curtosis', 'entropy', 'x0']].values
y = df['class'].values
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=13, test_size = 0.2)
```

```
weights = input("Enter 4 weights and 1 bias: ").split()
weights = [float(weight) for weight in weights]
weights = np.array(weights)
```

```
print('Momentum GD (Vanilla)')
train_vanilla(X_train,y_train,weights.copy())
```

```
Momentum GD (Vanilla)
Weights after 50 epochs: [-4.39170985 -2.71569597 -3.0187527  -0.86156677  2.25821314]

Error after 50 epochs: 0.006088524797520 56

Weights after 100 epochs: [-4.3363496  -2.3855687  -2.90810058 -0.5306964   3.26719043]

Error after 100 epochs: 0.003856002225942166

Weights after 150 epochs: [-4.28487923 -2.24673616 -2.81282294 -0.33574806  3.66350794]

Error after 150 epochs: 0.003475770487716895

Weights after 200 epochs: [-4.2521603  -2.19706772 -2.77880628 -0.25170574  3.86218869]

Error after 200 epochs: 0.0033891432638743723

Weights after 250 epochs: [-4.24795409 -2.18035801 -2.7705267  -0.21220897  3.96888747]

Error after 250 epochs: 0.0033664463651602423

Weights after 300 epochs: [-4.25985254 -2.17837356 -2.77476202 -0.19318412  4.03166804]

Error after 300 epochs: 0.0033586806182746218

Weights after 350 epochs: [-4.27904658 -2.18354698 -2.78513324 -0.1844168   4.07337171]

Error after 350 epochs: 0.0033545529592607644

Weights after 400 epochs: [-4.30131702 -2.19209819 -2.79833045 -0.18074011  4.10465489]

Error after 400 epochs: 0.0033514225794309322

Weights after 450 epochs: [-4.3246972  -2.20214642 -2.81269619 -0.17956236  4.13056985]

Error after 450 epochs: 0.0033486556753649322

Weights after 500 epochs: [-4.34826374 -2.21277075 -2.8274194  -0.17960206  4.15355833]

Error after 500 epochs: 0.003346090902787977
```

```
print('Momentum GD (Stochastic)')
train_stochastic(X_train,y_train,weights.copy())
```

```
Momentum GD (Stochastic)
C:\Users\Om Shete\AppData\Local\Temp\ipykernel_12340\4124679790.py:3: RuntimeWarning: overflow encountered in exp
  return (1 / (1 + np.exp(-lambdaa * net)))
Weights after 50 epochs: [-75.48036473 -50.36727033 -52.66577175  -5.4652907   52.97378194]

Error after 50 epochs: 0.009115767201358094

Weights after 100 epochs: [-75.46988016 -50.33777673 -52.69960101  -5.48295222  52.9809665 ]

Error after 100 epochs: 0.009115759107938713

Weights after 150 epochs: [-75.46003206 -50.31127291 -52.73012158  -5.49847125  52.98743587]

Error after 150 epochs: 0.009115752151093162

Weights after 200 epochs: [-75.44885613 -50.28209806 -52.76381042  -5.51532484  52.99456143]

Error after 200 epochs: 0.009115743364477184

Weights after 250 epochs: [-75.43385857 -50.24360928 -52.80832058  -5.53747081  53.00395412]

Error after 250 epochs: 0.009115727767399868
```

```
Weights after 300 epochs: [-75.40763212 -50.17673894 -52.88568705  -5.57615567  53.02023362]

Error after 300 epochs: 0.009115679917996871

Weights after 350 epochs: [-76.15748216 -48.07805044 -54.09633151 -11.74811181  56.47749927]

Error after 350 epochs: 0.006836649634338515

Weights after 400 epochs: [-78.88656793 -46.45063538 -53.06228899 -11.10489503  59.54460416]

Error after 400 epochs: 0.006836387673347274

Weights after 450 epochs: [-81.24619725 -45.793152   -51.18007778 -14.82990401  64.00280858]

Error after 450 epochs: 0.006844288173019668

Weights after 500 epochs: [-81.09465203 -45.80564705 -51.35447664 -14.68203285  64.08481472]

Error after 500 epochs: 0.006838394305651929
```

```
print('Momentum GD (Mini Batch)')
train_minibatch(X_train,y_train,weights.copy())
```

```
Momentum GD (Mini Batch)
Weights after 1 epochs: [-0.33463829 -0.39171181  0.4075694  -0.48639346  0.13469183]

Error after 1 epochs: 0.18023241371182047

Weights after 101 epochs: [-2.06093861 -1.15894496 -1.40096011 -0.07785838  2.29162845]

Error after 101 epochs: 0.0044560383270496945

Weights after 201 epochs: [-2.48459274 -1.35548264 -1.66769754 -0.0891239   2.65353428]

Error after 201 epochs: 0.00406856987571475

Weights after 301 epochs: [-2.77375195 -1.48673327 -1.84654205 -0.1014073   2.881557  ]

Error after 301 epochs: 0.0039017539823356716

Weights after 401 epochs: [-2.99681125 -1.58784422 -1.9843325  -0.11179156  3.05617612]

Error after 401 epochs: 0.0038032570740688535
```